

SQL Assignment – Data Modelling & Analysis

Schema Design (Star Schema)

Fact Table: fact_orders

Dimensions: dim_product, dim_customer, dim_location, dim_sales_rep, dim_date

Product Dimension

```
SELECT DISTINCT Item, Item_Category, SKU
INTO dim_product
FROM staging_data;
```

Location Dimension

```
SELECT DISTINCT Region, Country, City, State, Zip_Code, Warehouse, Store_ID
INTO dim_location
FROM staging_data;
```

Customer Dimension

```
SELECT DISTINCT Customer_ID, Customer_Name, Customer_Segment, Subscription
INTO dim_customer
FROM staging_data;
```

Fact Table

```
SELECT Order_ID, Customer_ID, Item, SKU, Rep, Store_ID, OrderDate,
Shipping_Date, Units, Unit_Cost, Total, Discount, COGS, Profit, Margin,
Tax, Tax_Amount, Order_Status, Sales_Channel, Payment_Method,
Return_Flag, CSAT_Score, Commission
INTO fact_orders
FROM staging_data;
```

Query 1: Revenue by Region

```
SELECT Region, Country, FORMAT(OrderDate, 'yyyy-MM') AS Month,
SUM(Total) AS Total_Revenue
FROM fact_orders f
JOIN dim_location l ON f.Store_ID = l.Store_ID
WHERE Order_Status = 'Completed'
GROUP BY Region, Country, FORMAT(OrderDate, 'yyyy-MM');
```

Query 2: Top 10 Items by Profit

```
SELECT TOP 10 Item, SUM(Total) Revenue, SUM(COGS) Total_COGS,
SUM(Profit) Total_Profit,
(SUM(Profit)*100.0/SUM(Total)) Profit_Margin_Percent
FROM fact_orders
GROUP BY Item
ORDER BY Total_Profit DESC;
```

Query 3: Return Rate

```
SELECT Customer_Segment, Sales_Channel,
COUNT(CASE WHEN Return_Flag=1 THEN 1 END)*1.0/COUNT(*) Return_Rate
FROM fact_orders f
JOIN dim_customer c ON f.Customer_ID=c.Customer_ID
GROUP BY Customer_Segment, Sales_Channel;
```

Query 4: CSAT

```
SELECT Rep, City, AVG(CSAT_Score) Avg_CSAT,
RANK() OVER(ORDER BY AVG(CSAT_Score) DESC) Rank_CSAT
FROM fact_orders f
JOIN dim_location l ON f.Store_ID=l.Store_ID
WHERE CSAT_Score IS NOT NULL
GROUP BY Rep, City;
```

Query 5: Discounts & Margins

```
WITH avg_discount AS (  
  SELECT AVG(Discount) avg_disc FROM fact_orders)  
SELECT Order_ID, Rep, Item, Discount, Margin, Total  
FROM fact_orders, avg_discount  
WHERE Discount > avg_disc AND Margin > 40;
```